

Solid-State Steerable Synthetic Jet Actuator via Phased-Array Acoustic Streaming

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Mac Mini M4 Compute Node, Independent Research

Abstract—This paper establishes the theoretical baseline and digital twin methodology for a solid-state synthetic jet actuator. Utilizing a 64-element uniform rectangular array (URA) of 40 kHz piezoelectric transducers, macroscopic fluid flow is generated entirely through the nonlinear attenuation of finite-amplitude acoustic waves. By coupling the Lighthill-Nyborg acoustic streaming equations with phased-array beamforming delay laws, we define a computational pipeline that extracts the time-averaged Reynolds stress tensor from a linear Helmholtz acoustic solution and applies it as a steady-state momentum source in the incompressible Navier-Stokes equations.

I. INTRODUCTION & SYSTEM ARCHITECTURE

Traditional fluid boundary-layer control systems, such as Dielectric Barrier Discharge (DBD) plasma actuators, suffer from high-voltage requirements, ozone generation, and environmental instability. We propose an alternative exploiting Eckart bulk acoustic streaming.

The physical hardware is constrained to an 8×8 planar array of 40 kHz transducers with a pitch of $d = 10.5$ mm. Driven at $V_{DD} = 24$ V, the system acts as a solid-state synthetic jet. Because the array pitch exceeds the spatial Nyquist criterion ($\lambda/2 \approx 4.28$ mm), grating lobes are unavoidable, necessitating strict spatial targeting. Empirical baseline measurements suggest maximum achievable time-averaged flow velocities on the order of 0.1 m/s to 0.5 m/s at high Sound Pressure Levels (SPL).

II. PHASED-ARRAY FOCUSING MATHEMATICS

To synthesize a focused momentum jet at an arbitrary spatial coordinate $\mathbf{r}_f = (x_f, y_f, z_f)$, the acoustic wavefronts from all N elements must arrive perfectly in phase. Let the position of the i -th transducer be $\mathbf{r}_i = (x_i, y_i, 0)$.

The geometric distance from element i to the focal point is:

$$R_i = \|\mathbf{r}_i - \mathbf{r}_f\| = \sqrt{(x_i - x_f)^2 + (y_i - y_f)^2 + z_f^2} \quad (1)$$

To ensure simultaneous wavefront arrival, we define the time-delay law Δt_i relative to the element with the shortest path:

$$\Delta t_i = \frac{R_i}{c_0} - \frac{\min_j R_j}{c_0} \quad (2)$$

where $c_0 \approx 343$ m/s is the local speed of sound in air.

Due to spatial aliasing ($d > \lambda/2$), parasitic intensity maxima (grating lobes) will appear at angles θ_k satisfying:

$$\sin \theta_k = \frac{k\lambda}{d} - \sin \theta, \quad k = \pm 1, \pm 2, \dots \quad (3)$$

These lobes will consume a fraction of the acoustic energy, implicitly bounding the maximum magnitude of the resulting streaming velocity.

III. ACOUSTIC STREAMING THEORY

The generation of macroscopic wind from acoustic waves is a second-order nonlinear phenomenon. We decouple the physics using perturbation theory, where the instantaneous velocity $\mathbf{u}$ and density ρ are expanded as:

$$\mathbf{u} = \epsilon \mathbf{u}_1 + \epsilon^2 \mathbf{u}_2 \quad (4)$$

$$\rho = \rho_0 + \epsilon \rho_1 + \epsilon^2 \rho_2 \quad (5)$$

where subscript 1 denotes the first-order (40 kHz) acoustic field, and subscript 2 denotes the steady (DC) fluid flow.

As derived by Nyborg and Lighthill, time-averaging the compressible Navier-Stokes equations over one acoustic period reveals that the first-order momentum flux leaves a non-zero residual force. The time-averaged second-order momentum equation behaves as a Stokes flow driven by a volumetric body force density $\mathbf{F}_{\text{rad}}$:

$$\mu \nabla^2 \langle \mathbf{u}_2 \rangle - \nabla \langle p_2 \rangle + \mathbf{F}_{\text{rad}} = 0 \quad (6)$$

The acoustic radiation force $\mathbf{F}_{\text{rad}}$ is defined by the spatial divergence of the Reynolds stress tensor of the first-order velocity field:

$$F_{\text{rad},i} = -\frac{\partial}{\partial x_j} \langle \rho_0 u_{1,i} u_{1,j} \rangle \quad (7)$$

IV. CONCLUSION & NEXT STEPS

With the theoretical framework defined, the next phase bridges the analytical mathematics to finite-volume simulations. The time-delay equations (Eq. 2) will inform the boundary conditions of a linear Helmholtz solver to yield $\mathbf{u}_1$. Subsequently, Eq. 6 will be computed discretely to generate $\mathbf{F}_{\text{rad}}$, serving as the momentum source term in an incompressible Navier-Stokes solver.